MATLAB Vibrations Lab Plotting Scripts

Chat transcript, 21–27 October 2026 (long code listings replaced with "code output")

You
(Attached: Vibrations_datasets.zip.) Set up MATLAB code to load data from the "MAE Vibrations" folder, go into the "Vibrations datasets" zip, then the M003 folder. Files are named like "10_8_2026m003_hammer_x_ydat.txt", "..._x_yfft.txt" and "Shakerx_inch_y.txt" (x = 16, 20 or 24 in; y = a/b). Let the user type only the length and experiment. Make an amplitude vs. frequency plot with uncertainties and error bars, plus a theoretical curve: R times omega^2, scaled by the ratio of the measured maximum to that curve's maximum.

Claude
Wrote **plotVibration.m**, called as plotVibration('dat'|'fft'|'shaker', 16|20|24, 'a'|'b').

- Finds the folder, extracts the zip the first time, and locates the file by pattern.
- Hammer "dat" files are frequency data and "fft" files are time series (an FFT is taken). Column 2 is the beam output, column 3 the hammer input.
- Theory R = 1/sqrt((1-r^2)^2+(2 zeta r)^2), times omega^2, scaled to the measured peak.
- Asked for: uncertainty values, the theory formula, shaker units. Flagged the 20 in B run as odd.

You
(Attached: Vibrations_Lab_Procedure.pdf.) This should have all the necessary formulas and values. For any missing values use standard published values and cite the website.

Claude
Rebuilt the theory around the procedure:

- fn from beam theory: k = 3EI/L^3, equivalent mass 33/140 of the beam mass (eqs 3, 4), with propagated uncertainty.
- zeta from log decrement of the hammer time series (eqs 5, 6). Measured fn via eq 7 (hammer) and eq 8 (shaker).
- Cited values: MatWeb AISI 1018 (E = 200 GPa, 7.87 g/cc), Total Materia (ranges), NI-9234 specs (0.34 % gain error), McMaster bar tolerances, US Patent 6,526,835 (0.2357 M), McGill ECSE 304 notes (magnitude ratio).
- Noted measured fn is about 8–13 % below theory (clamp compliance).

You
make the code copy and pasteable

Claude
Turned it into a plain script (plotVibrationData.m) with the file type, length and experiment set at the top. Pasted the full script.

[code output]

You
Now I need Amplitude vs Normalized Frequency plots for the output for each length of beam, with uncertainties. Same plot, but the x-axis divided by the theoretical undamped natural frequency.

Claude
Added normalizeFreq (x = f/fn,theory, with fn uncertainty in the horizontal error bars) and a loop over all three lengths.

[code output]

You
I want it to function like the last code, one plot at a time. The beam is 2.07 lb, 0.9985 in wide, 29.5 in long, 0.288 in thick, carbon steel.

Claude

Back to one plot per run, with the measured beam and the mass of the vibrating part = (L/29.5) x 2.07 lb.

- New theory fn: 39.50 ± 0.99, 25.28 ± 0.63, 17.56 ± 0.44 Hz.
- Flagged: a 29.5 x 0.9985 x 0.288 in steel bar should weigh about 2.41 lb; 2.07 lb fits 0.25 in thickness.

[code output]

You
Now I need a plot of undamped natural and resonance frequencies for my various lengths (with error bars), using the experimental damping ratio from the hammer test, with uncertainties.

Claude
New script plotNaturalFrequencies.m: theory, hammer and shaker fn and fr vs. length with propagated uncertainties, plus a panel for fn – fr (under 1 mHz because zeta is about 0.002–0.003).

You
make it copy and pasteable always

Claude
Pasted the full script, and will paste code in every reply from now on.

[code output]

You
does this plot all three lengths at the same time? i need it to

Claude
Yes. One run plots 16, 20 and 24 in on one figure.

You
i need the plot to have error bars and i need the plot to read fr and fn

Claude
Split into labelled f_n and f_r panels with bigger error bars and a value ± uncertainty label on every point.

[code output]

You
it needs to be one plot

Claude
Put everything on one axes: filled markers for fn, open markers for fr, side by side at each length, with error bars and labels.

[code output]

You
Now I need a code that makes 4 subplots in one figure: voltage vs time (input and output) and amplitude vs frequency (input and output). Only one length, typed as 16, 20 or 24.

Claude
New script plotHammerSignals.m: a 2 x 2 figure with the hammer strike and the output peak labelled.

[code output]

You
Now 2 subplots in 1 figure: magnitude of frequency response |H(f)| vs frequency and phase lag vs frequency. y-axis labels |H(f)| and Phase lag. Let the user pick the hammer length.

Claude
New script plotFrequencyResponse.m: H = FFT(output)/FFT(input) from the hammer time series, with an SDOF theory overlay.

- The phase is shown for displacement by default (0 → 90 → 180 deg); the raw accelerometer phase is optional.

  [code output]

You
Now source data from the "Lab Data V" folder: files "10_8_2026jamesXvY.txt", X = experiment 1–14, Y = dat or fft. I need 1 figure per parameter change (length, damping coefficient, end mass, material), each with |H(f)| and phase lag subplots.

Claude
Asked for the files and the run-to-parameter mapping (proposed runs 1–3 length, 4–8 damping, 9–11 end mass, 12–14 material).

You
(Attached: Lab_Data_V.zip.) Everything you said about the runs is correct.

Claude
New script plotSimulationResponse.m: "fft" holds |H(f)| and "dat" holds the phase angle. One figure per parameter, or customRuns.

- Flagged recording problems: peak heights match theory but several runs' frequency axes match the previous run (the simulation must be run twice per setting); run 3 looks like c = 9 kg/s; runs 4–8 resonate at 1.58 Hz.

  [code output]

You
Instead of run on the plot I need it to say the length used or the material type etc.

Claude
Legends now show the case (L = 16 in, c = 1 kg/s, m_end = 0.25 kg, Aluminum).

[code output]

You
(Attached: undampednatresonance.m, twosubplotqfive.m, ShakerTestCode.m, shakernormalizedfreq.m, hammerquadplot.m.) Put all of these together without losing any functionality, and make the code as short as possible.

Claude
Combined them into **vibrationLab.m** (479 lines, compared with 2,429 for the five scripts). plotType selects 'amplitude', 'natres', 'frf' or 'quad'. Shared code is written once and every setting is kept. The measured beam values are used; the old 1.00 x 0.25 in values can be restored in the settings.

[code output]

You
(Attached: part6plots.m.) Merge that code with this code, keeping the function the same and the code short.

Claude
Added the simulation plots as plotType = 'sim' (569 lines in total). The sim frequency limit is now simFreqLim, and 'sim' only needs the Lab Data V folder.

[code output]

You
make this chat a pdf. any areas with extremely long code just say "code output"

Claude
Created this PDF.